$$\because \alpha > 0, \quad \beta > 0, \quad q > 0, \quad \beta - \alpha > q$$

$$\therefore (\beta - \alpha)^2 > q^2$$

$$\beta^2 + \alpha^2 - 2\alpha\beta - q^2 > 0$$

$$\frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta} - 2 > 0$$

$$u_{n+2} = \frac{u_{n+1}^2 - q^2}{u_n}, \qquad u_{n-1} = \frac{u_n^2 - q^2}{u_{n+1}}$$

$$\begin{aligned} \text{LHS} &= u_n(u_n + u_{n+2}) \\ &= u_n\left(u_n + \frac{u_{n+1}^2 - q^2}{u_n}\right) \\ &= u_n^2 + u_{n+1}^2 - q^2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= u_{n+1}(u_{n-1} + u_{n+1}) \\ &= u_{n+1}\left(\frac{u_n^2 - q^2}{u_{n+1}} + u_{n+1}\right) \\ &= u_n^2 + u_{n+1}^2 - q^2 \end{aligned}$$

$$\therefore \boxed{\text{LHS} = \text{RHS}}$$

$$u_n(u_n + u_{n+2}) = u_{n+1}(u_{n-1} + u_{n+1})$$

$$\implies \frac{u_{n+2} + u_n}{u_{n+1}} = \frac{u_{n+1} + u_{n-1}}{u_n} = \text{constant}$$

Let $p = \dfrac{u_{n+1} + u_{n-1}}{u_n}$

$$p = \frac{u_2 + u_0}{u_1} = \frac{\frac{\beta^2 - q^2}{\alpha} + \alpha}{\beta} = \frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta}$$

$$u_{n+1} - pu_n + u_{n-1} = 0$$

$$p = \frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta} > 2$$

$$u_{n+1} - u_n = pu_n - u_{n-1} - u_n = (p-2)u_n + (u_n - u_{n-1})$$

$$\because u_1 - u_0 = \beta - \alpha > q > 0$$

$$\therefore u_{n+1} - u_n > u_n - u_{n-1} > 0$$

When $\beta = \alpha + q$,

$$p = \frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta} = 2$$

$$u_{n+1} - 2u_n + u_{n-1} = 0$$

$$\implies u_{n+1} - u_n = u_n - u_{n-1}$$

$$\therefore u_{n+1} - u_n = u_1 - u_0 = q > 0$$

The sequence is strictly increasing.

This solution was submitted by **Jon**
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